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**RadioDNS Hybrid Radio;**
**Hybrid lookup for radio services**

EBU

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# Foreword

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NOTE: The EBU/ETSI JTC Broadcast was established in 1990 to co-ordinate the drafting of standards in the specific field of broadcasting and related fields. Since 1995 the JTC Broadcast became a tripartite body by including in the Memorandum of Understanding also CENELEC, which is responsible for the standardization of radio and television receivers. The EBU is a professional association of broadcasting organizations whose work includes the co-ordination of its members' activities in the technical, legal, programme-making and programme-exchange domains. The EBU has active members in about 60 countries in the European broadcasting area; its headquarters is in Geneva.

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# 1 Scope

The present document defines the methodology for discovering an Authoritative FQDN for a radio service, including discovery using DNS queries to radiodns.org, a root domain name server operated by RadioDNS. The present document also defines the construction of a unique ServiceIdentifier parameter and bearerURI for a radio service.

This version includes the addition of client identification.

NOTE: Specifications for applications built upon the RadioDNS methodology can be found at http://radiodns.org/developers/documentation/.

# 2 References

## 2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference/.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI EN 300 401: "Radio Broadcasting Systems; Digital Audio Broadcasting (DAB) to mobile, portable and fixed receivers".

[2] ETSI ES 201 980: "Digital Radio Mondiale (DRM); System Specification".

[3] ETSI TS 102 386: "Digital Radio Mondiale (DRM); AM signalling system (AMSS)".

[4] National Radio Systems Committee NRSC-5-B:2008: "In-band/on-channel Digital Radio Broadcasting Standard".

[5] ETSI TS 102 818: "Digital Audio Broadcasting (DAB); Digital Radio Mondiale (DRM); XML Specification for Electronic Programme Guide (EPG)".

[6] IETF RFC 1035 (1987): "Domain Names - Implementation and Specification".

[7] IETF RFC 3761 (2004): "The E.164 to Uniform Resource Identifiers (URI) Dynamic Delegation Discovery System (DDDS) Application (ENUM)".

[8] IEC 62106:2015: "Specification of the Radio Data System (RDS) for VHF/FM sound broadcasting in the frequency range from 87,5 MHz to 108,0 MHz".

[9] National Radio Systems Committee NRSC-4-B: "Specification of the radio broadcast data system (RBDS)".

NOTE: Available at https://www.nrscstandards.org/standards-and-guidelines/documents/standards/nrsc-4-b.pdf

[10] ISO 3166-1: "Codes for the representation of names of countries and their subdivisions - Part 1: Country codes".

[11] Void.

[12] ETSI TS 101 756: "Digital Audio Broadcasting (DAB); Registered Tables".

[13] IETF RFC 2818: "HTTP over TLS".

[14] IETF RFC 7235: "Hypertext Transfer Protocol (HTTP/1.1): Authentication".

[15] SY-IDD-1020s: "HD Radio™ Air Interface Design Description, Station Information, Service Transport".

## 2.2 Informative references

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The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

Not applicable.

# 3 Definition of terms, symbols and abbreviations

## 3.1 Terms

For the purposes of the present document, the following terms apply:

**application:** means of enhancing an associated radio service, using an IP connection, with additional content, functionality, or interactivity

**authoritative FQDN:** internet domain for a service provider

**bearer:** method of carriage of the service

**bearerURI:** unique identifier for the service to be used in SPI SI documents

**char:** single character

**client identifier:** non-unique identifier sent by a client to a service provider

**hexadecimal:** representation of a number in base-16 using the characters 0-9, a-f

**nibble:** four-bit aggregation, or half an octet

**RadioDNS FQDN:** internet domain constructed only for the purposes of querying DNS

**service:** radio service or data service

**ServiceIdentifier:** string that uniquely identifies a radio service within the scope of an Authoritative FQDN

**service provider:** organization providing RadioDNS Hybrid Radio applications

**string:** zero or more characters in the range 0-9, a-z

## 3.2 Symbols

Void.

## 3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

| | |
|---|---|
| AM | Amplitude Modulation |
| AMSS | Amplitude Modulation Signalling System |
| ASF | Advanced Systems Format |
| CNAME | DNS Canonical NAME record |
| DAB | Digital Audio Broadcasting |
| DNS | Domain Name System |
| DRM | Digital Radio Mondiale |
| ECC | Extended Country Code |
| EId | Ensemble Identifier |
| FCC | Federal Communications Commission |
| FIG | Fast Information Group |
| FM | Frequency Modulation |
| FQDN | Fully Qualified Domain Name |
| GCC | Global Country Code |
| HTTP | HyperText Transfer Protocol |
| IBOC | In-Band On-Channel |
| IP | Internet Protocol |
| PI | Programme Identification |
| RBDS | Radio Broadcast Data System |
| RDS | Radio Data System |
| SCIdS | Service Component Identifier within a Service |
| SI | Service Information |
| SId | Service Identifier |
| SPI | Service and Programme Information |
| SPS | Supplemental Program Service |
| SRV | DNS SeRVice record |
| TLS | Transport Layer Security |
| TTL | Time To Live |
| URI | Uniform Resource Identifier |
| URL | Uniform Resource Locator |
| VHF | Very High Frequency |

# 4 Introduction

It is possible to supplement uni-directional radio services with applications that can take advantage of bi-directional communication using the IP protocol. These applications may enhance the radio services with which they are associated with additional content or functionality, or enable interactivity.

Radio devices should be aware of what IP delivered applications are available for each radio service it receives. Standardizing the methodology to locate these applications allows a manufacturer to support IP delivered applications directly on the device.

The present document standardizes the methodology for locating the Authoritative Fully Qualified Domain Name (FQDN) for radio services using the following radio systems: FM with RDS [8] or RBDS [9], DAB/DAB+ [1], DRM [2], AM with AMSS [3], and IBOC [4].

The present document standardizes a methodology to locate applications based upon the existing DNS methodology [6]. A RadioDNS FQDN is created from known broadcast parameters, and DNS is used to resolve this RadioDNS FQDN to a CNAME record containing the Authoritative FQDN for the service provider.

The basis for this methodology broadly follows that used to map E.164 format telephone numbers to domains [7].

The present document also standardizes how to locate the Authoritative FQDN without the use of DNS lookup.

The Authoritative FQDN for a service can be acquired through a series of processes, shown in figure 1.

The present document also standardizes how devices identify themselves when connecting to service providers.

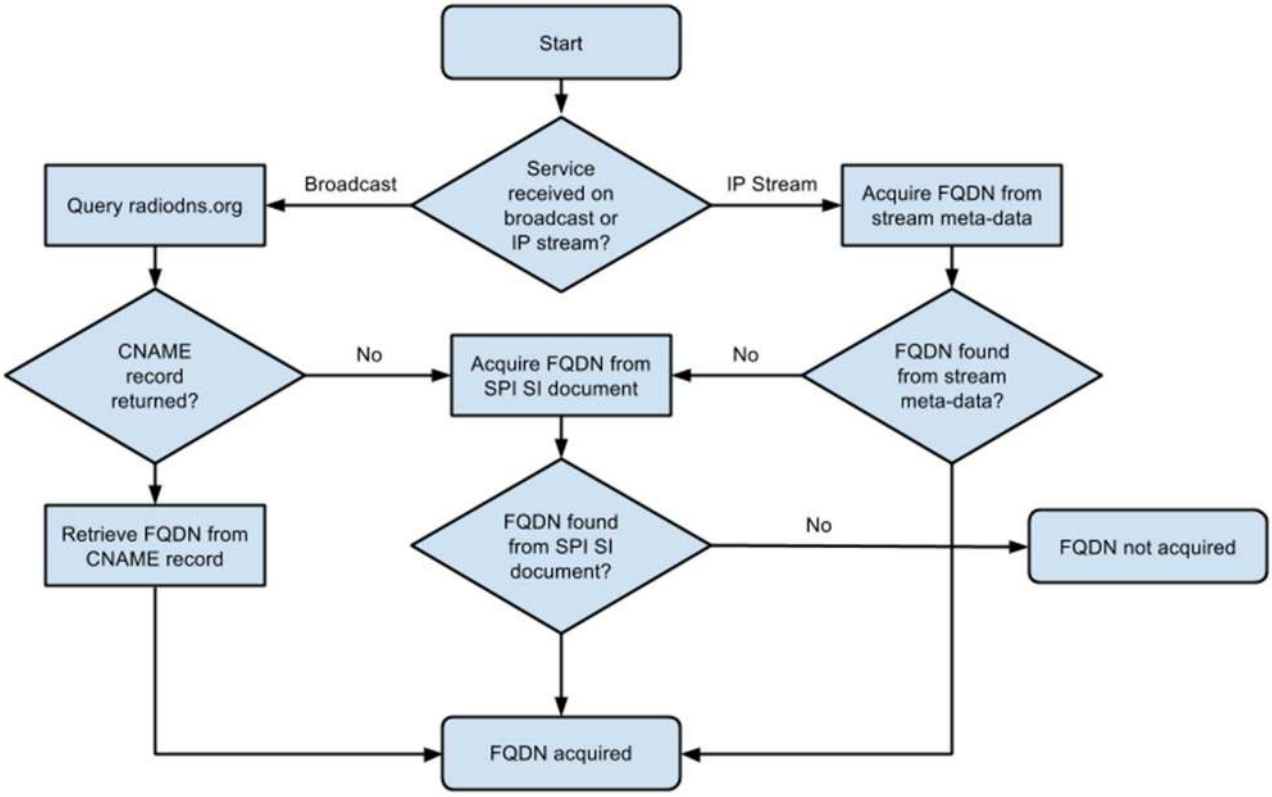



**Figure 1: Process to acquire Authoritative FQDN for a service**

The service is also given a ServiceIdentifier parameter, which is unique within the scope of an Authoritative FQDN.

The service may also be given a bearerURI parameter, which allows location of the service when placed in an SPI SI document.

Clause 5 describes how to resolve the Authoritative FQDN and construct the ServiceIdentifier and bearerURI for broadcast radio services.

Clause 6 describes how to resolve the Authoritative FQDN and ServiceIdentifier and construct the bearerURI for streaming radio services.

Clause 7 describes how to resolve the Authoritative FQDN and ServiceIdentifier from an SPI SI document.

Methods for discovery of the SPI SI document are defined in ETSI TS 102 818 [5].

# 5 Authoritative FQDN resolution, and ServiceIdentifier and bearerURI construction for broadcast services

## 5.1 RadioDNS FQDN, ServiceIdentifier and bearerURI construction

### 5.1.1 FM with RDS/RBDS

#### 5.1.1.1 RDS/RBDS parameters

The FM system supports identification of a radio service through transmission of meta-data by using RDS [8] or RBDS [9].

The parameters are defined in table 1.

**Table 1: RDS/RBDS parameter description**

| Parameter | Description | Value | Status |
|---|---|---|---|
| gcc | The Global Country Code (GCC) of the country of origin of the service (see annex A). | 3-char hexadecimal | mandatory |
| pi | Received RDS/RBDS Programme Identification (PI) code. | 4-char hexadecimal | mandatory |
| frequency | Frequency on which the service broadcast is received, formatted to 5 characters in units of 100 KHz. Frequencies below 100 MHz shall be supplied with a leading zero, for example 95,8 MHz would be represented as "09580", 104,9 MHz as "10490". | 5-char string | mandatory |

NOTE: During the development of RadioDNS, it was permitted to compile the RadioDNS FQDN using the ISO 3166-1 [10] alpha-2 country code as an alternative to the GCC. However, since the GCC can be derived from location information and the PI code, only the GCC has been standardized.

#### 5.1.1.2 Construction of RadioDNS FQDN

The RadioDNS FQDN for a VHF/FM service is compiled as follows:

```
<frequency>.<pi>.<gcc>.fm.radiodns.org
```

Some examples of RadioDNS FQDNs constructed from broadcast parameters are shown in table 2.

**Table 2: Example of RadioDNS FQDN construction for RDS/RBDS**

| GCC | PI | Frequency (MHz) | RadioDNS FQDN |
|---|---|---|---|
| ce1 | c586 | 95,8 | 09580.c586.ce1.fm.radiodns.org |
| de0 | d1e0 | 103,9 | 10390.d1e0.de0.fm.radiodns.org |

#### 5.1.1.3 Construction of ServiceIdentifier

The ServiceIdentifier for a VHF/FM service is compiled as follows:

```
fm/<gcc>/<pi>/<frequency>
```

Some examples of ServiceIdentifiers constructed from broadcast parameters are shown in table 3.

**Table 3: Example of RadioDNS ServiceIdentifier construction for RDS/RBDS**

| GCC | PI | Frequency (MHz) | RadioDNS ServiceIdentifier |
|---|---|---|---|
| ce1 | c586 | 95,8 | fm/ce1/c586/09580 |
| de0 | d1e0 | 103,9 | fm/de0/d1e0/10390 |

#### 5.1.1.4 Construction of bearerURI

The bearerURI for a VHF/FM service is compiled as follows:

```
fm:<gcc>.<pi>.<frequency>
```

The <frequency> element may be replaced by the asterisk ("*") character to signify any frequency. In this case the PI code alone shall be used by the device to locate the source.

Some examples of FM bearerURIs constructed from broadcast parameters are shown in table 4.

**Table 4: Example of RadioDNS bearerURI construction for RDS/RBDS**

| GCC | PI | Frequency (MHz) | RadioDNS bearerURI |
|---|---|---|---|
| ce1 | c586 | 95,8 | fm:ce1.c586.09580 |
| de0 | d1e0 | 103,9 | fm:de0.d1e0.10390 |
| ce1 | c201 | many | fm:ce1.c201.* |

## 5.1.2 Digital Audio Broadcasting (DAB/DAB+)

### 5.1.2.1 DAB/DAB+ parameters

The parameters are defined in table 5.

**Table 5: DAB parameter description**

| Parameters | Description | Value | Status |
|---|---|---|---|
| gcc | The Global Country Code (GCC) of the country of origin of the service (see annex A) | 3-char hexadecimal | mandatory |
| eid | The Ensemble Identifier (EId) of the service | 4-char hexadecimal | mandatory |
| sid | The Service Identifier (SId) of the service | 4- or 8-char hexadecimal | mandatory |
| scids | The Service Component Identifier within the Service (SCIdS) of the service component | 1-char hexadecimal | mandatory |
| uatype | The User Application Type (UAtype) of the data component | 3-char hexadecimal | mandatory for data components, otherwise omitted |

For data services (or data components of audio services) the **uatype** parameter is also mandatory.

### 5.1.2.2 Construction of RadioDNS FQDN

The RadioDNS FQDN for a DAB/DAB+ service is compiled as follows:

```
[<uatype>.]<scids>.<sid>.<eid>.<gcc>.dab.radiodns.org
```

Some examples of RadioDNS FQDNs constructed from broadcast parameters are shown in table 6.

**Table 6: Example of RadioDNS FQDN construction for DAB**

| GCC | EId | SId | SCIdS | UAType | RadioDNS FQDN |
|---|---|---|---|---|---|
| de0 | 100c | d220 | 0 | | 0.d220.100c.de0.dab.radiodns.org |
| ce1 | c18c | cc86 | 0 | | 0.cc86.c18c.ce1.dab.radiodns.org |
| ce1 | c185 | e1c00098 | 0 | 004 | 004.0.e1c00098.c185.ce1.dab.radiodns.org |

### 5.1.2.3 Construction of ServiceIdentifier

The ServiceIdentifier for a DAB/DAB+ service is compiled as follows:

```
dab/<gcc>/<eid>/<sid>/<scids>[/<uatype>]
```

The <separator> element is application specific. The inclusion of <uatype> is mandatory for data services or data components of audio services.

Some examples of ServiceIdentifiers constructed from broadcast parameters are shown in table 7.

**Table 7: Example of RadioDNS ServiceIdentifer construction for DAB**

| GCC | EId | SId | SCIdS | UAType | RadioDNS ServiceIdentifier |
|---|---|---|---|---|---|
| de0 | 100c | d220 | 0 | | dab/de0/100c/d220/0 |
| ce1 | c18c | cc86 | 0 | | dab/ce1/c18c/cc86/0 |
| ce1 | c185 | e1c00098 | 0 | 004 | dab/ce1/c185/e1c00098/0/004 |

#### 5.1.2.4 Construction of bearerURI

The bearerURI for a DAB/DAB+ service is compiled as follows:

```
dab:<gcc>.<eid>.<sid>.<scids>[.<uatype>]
```

The inclusion of `<uatype>` is mandatory for data services or data components of audio services.

Some examples of bearerURIs constructed from broadcast parameters are shown in table 8.

**Table 8: Example of RadioDNS bearerURI construction for DAB**

| GCC | EId | SId | SCIdS | UAType | RadioDNS bearerURI |
|---|---|---|---|---|---|
| de0 | 100c | d220 | 0 | | dab:de0.100c.d220.0 |
| ce1 | c18c | cc86 | 0 | | dab:ce1.c18c.cc86.0 |
| ce1 | c185 | e1c00098 | 0 | 004 | dab:ce1.c185.e1c00098.0.004 |

### 5.1.3 Digital Radio Mondiale (DRM)

#### 5.1.3.1 DRM parameters

The parameters are defined in table 9.

**Table 9: DRM parameter description**

| Parameters | Description | Value | Status |
|---|---|---|---|
| sid | The Service Identifier (SId) of the service | 6-char hexadecimal | mandatory |
| appdomain | The application domain of the data component | 1-char hexadecimal | mandatory for data components, otherwise omitted |
| uatype | The user application type of the data component | 3-char hexadecimal | mandatory for data components, otherwise omitted |

The SId value for DRM is intended to be suitably unique internationally so as to not require region identification.

#### 5.1.3.2 Construction of RadioDNS FQDN

The RadioDNS FQDN for a Digital Radio Mondiale service is compiled as follows:

```
[<uatype>.<appdomain>]<sid>.drm.radiodns.org
```

Some examples of RadioDNS FQDNs constructed from broadcast parameters are shown in table 10.

**Table 10: Example of RadioDNS FQDN construction for DRM**

| SId | App Domain | UAType | RadioDNS FQDN |
|---|---|---|---|
| e1c238 | | | e1c238.drm.radiodns.org |
| f07256 | 1 | 00d | 00d.1.f07256.drm.radiodns.org |
| a13002 | | | a13002.drm.radiodns.org |

#### 5.1.3.3 Construction of ServiceIdentifier

The ServiceIdentifier for a Digital Radio Mondiale service compiled as follows:

```
drm/<sid>[/<appdomain>/<uatype>]
```

Some examples of ServiceIdentifiers constructed from broadcast parameters are shown in table 11.

**Table 11: Example of RadioDNS ServiceIdentifer construction for DRM**

| SId | App Domain | UAType | RadioDNS ServiceIdentifer |
|---|---|---|---|
| e1c238 | | | drm/e1c238 |
| f07256 | 1 | 00d | drm/f07256/1/00d |
| a13002 | | | drm/a13002 |

#### 5.1.3.4 Construction of bearerURI

The bearerURI for a Digital Radio Mondiale service is compiled as follows:

```
drm:<sid>[.<appdomain>.<uatype>]
```

Some examples of bearerURIs constructed from broadcast parameters are shown in table 12.

**Table 12: Example of RadioDNS bearerURI construction for DRM**

| SId | App Domain | UAType | RadioDNS bearerURI |
|---|---|---|---|
| e1c238 | | | drm:e1c238 |
| f07256 | 1 | 00d | drm:f07256.1.00d |
| a13002 | | | drm:a13002 |

### 5.1.4 AM Signalling System (AMSS)

#### 5.1.4.1 AMSS parameters

The parameters are defined in table 13.

**Table 13: AMSS parameter description**

| Parameters | Description | Value | Status |
|---|---|---|---|
| sid | The Service Identifier (SId) of the service | 6-char hexadecimal | mandatory |

The SId value for AMSS is intended to be suitably unique internationally so as to not require region identification.

#### 5.1.4.2 Construction of RadioDNS FQDN

The RadioDNS FQDN for an AM service with AMSS is compiled as follows:

```
<sid>.amss.radiodns.org
```

#### 5.1.4.3 Construction of ServiceIdentifier

The ServiceIdentifier for an AM service with AMSS is compiled as follows:

```
amss/<sid>
```

#### 5.1.4.4 Construction of bearerURI

The bearerURI for an AM service with AMSS is compiled as follows:

```
amss:<sid>
```

## 5.1.5 IBOC

### 5.1.5.1 IBOC parameters

The parameters are defined in table 14.

**Table 14: IBOC parameter description**

| Parameters | Description | Value | Status |
|---|---|---|---|
| `tx` | Transmitter Identifier<br>Service broadcast identifier | 5-char hexadecimal | mandatory |
| `cc` | Country Code<br>Service broadcast country code | 3-char hexadecimal | mandatory |
| `mId` | The identifier for a multicast Supplemental Program Service (SPS) channel | 1-char hexadecimal | mandatory for supplemental services, otherwise omitted |

The `tx` value is the FCC facility code of the transmitter in the United States of America, or a unique identifier supplied by the licensors of the IBOC technology in other places.

The `cc` value is the hexadecimal representation of the encoded country ID [15].

EXAMPLE: the value of `cc` for the United States of America is `0x292`.

The `mId` is only defined if the bearer describes a Supplemental Program Service (SPS), defined as a supplemental "multicast" service to the main service. For a bearer describing the Main Program Service (HD-1), this value is omitted. For a bearer describing the first SPS channel (HD-2), `mId` shall be 2, for the second SPS channel (HD-3), `mId` shall be 3, and so on.

### 5.1.5.2 Construction of RadioDNS FQDN

The RadioDNS FQDN for an IBOC service is compiled as follows:

```
[<mId>.]<tx>.<cc>.hd.radiodns.org
```

**Table 15: Example of RadioDNS FQDN construction for IBOC**

| tx | cc | mId | RadioDNS FQDN |
|---|---|---|---|
| 07426 | 292 | | 07426.292.hd.radiodns.org |
| 07426 | 292 | 2 | 2.07426.292.hd.radiodns.org |

### 5.1.5.3 Construction of ServiceIdentifier

The ServiceIdentifier for an IBOC service is compiled as follows:

```
hd/<cc>/<tx>[/<mId>]
```

**Table 16: Example of RadioDNS ServiceIdentifier construction for IBOC**

| tx | cc | mId | RadioDNS ServiceIdentifier |
|---|---|---|---|
| 07426 | 292 | | hd/292/07426 |
| 07426 | 292 | 2 | hd/292/07426/2 |

### 5.1.5.4 Construction of bearerURI

The bearerURI for an IBOC service is compiled as follows:

```
hd:<cc>.<tx>[.<mId>]
```

**Table 17: Example of RadioDNS bearerURI construction for IBOC**

| tx | cc | mId | RadioDNS bearerURI |
|---|---|---|---|
| 07426 | 292 | | hd:292.07426 |
| 07426 | 292 | 2 | hd:292.07426.2 |

## 5.2 Resolution of Authoritative FQDN

The RadioDNS FQDN, constructed from the broadcast parameters, is used to acquire the Authoritative FQDN. Making a DNS query with a RadioDNS FQDN will return a single CNAME record containing the Authoritative FQDN of the service provider. If no CNAME is returned, then the service has not been registered.

EXAMPLE: Consider an FM service identified by the RadioDNS FQDN:

```
09580.c479.ce1.fm.radiodns.org
```

Using the nslookup tool would yield the following lookup result:

```
canonical name = rdns.musicradio.com
```

Therefore, for this service, the Authoritative FQDN is:

```
rdns.musicradio.com
```

The broadcast parameters should be continuously monitored. If any broadcast parameter changes (for example, a change to the RDS/RBDS PI code), the process of resolving the Authoritative FQDN should be repeated using the new broadcast parameters.

The TTL (Time To Live) parameters of the Authoritative FQDN shall be queried and respected.

Upon expiry of the TTL, the process of resolving the Authoritative FQDN shall be repeated.

If the Authoritative FQDN has changed, then all active applications shall be notified and each application shall repeat its own process for connecting to resources using the updated Authoritative FQDN.

# 6 Authoritative FQDN and ServiceIdentifier resolution and bearerURI construction for IP-streamed services

## 6.1 General

An Authoritative FQDN may also be provided for IP-streamed services, by sending the value as part of the in-stream metadata of the IP stream. This is defined as the parameter **fqdn**.

Since no broadcast parameters exist for such services, an additional parameter is required to provide disambiguation so that the particular RadioDNS application can determine the exact service being used. This is defined as the parameter **sid**.

The **sid** shall be unique across all services using the same Authoritative FQDN for application discovery, with a maximum character limit of 16 characters in the range [a-z][0-9]. The exact use of this parameter is specific to the RadioDNS application being used.

For any streaming protocol where the **fqdn** and **sid** parameters are sent as in-stream metadata at regular intervals, the values shall be monitored after they have been initially acquired. If these values are found to change at any point, the old values will be deemed to have expired, and the process of resolving the Authoritative FQDN shall be repeated.

If the Authoritative FQDN has changed, then all active applications shall be notified and each application shall repeat its own process for connecting to resources using the updated Authoritative FQDN.

## 6.2 Inclusion of parameters into stream metadata

### 6.2.1 Streaming transports

#### 6.2.1.1 SHOUTcast

SHOUTcast uses a client-server model, with each component communicating via a network protocol that intermingles audio or video data with metadata such as song titles and the station name. It uses HTTP as a transport protocol.

NOTE: Additional information is available from https://forums.radiotoolbox.com/viewtopic.php?t=74.

The parameters should be contained within the initial HTTP Response at the start of the stream, using the HTTP response header **icy-url**, which has a defined usage within the SHOUTcast specification. Its value should be of the form:

```
http://<fqdn>/<sid>
```

If a Service Provider wishes to also support the intended functionality of this parameter to provide a URL to a website, it is recommended that HTTP requests to this URL are handled appropriately (such as delivering a web-page, or returning an HTTP 302 response to re-direct the browser to an alternative URL).

#### 6.2.1.2 ASF

Advanced Systems Format (ASF) is a container format that is part of the Windows Media framework. It typically defines a payload containing multiple streams of data, e.g. audio and a metadata stream.

An additional stream shall be created, solely containing the **fqdn** and **sid**, declared as Custom Metadata using key/value pairs for attributes with the following keys:

- **radiodns-fqdn** for the **fqdn**
- **radiodns-sid** for the **sid**

It is recommended that the values be programmatically specified as a null-terminated Unicode string, using the default platform language.

NOTE: If using Windows Media Encoder, this can be entered in as Custom Metadata when setting up the stream.

#### 6.2.1.3 Flash Audio

Flash Audio is a container format for audio and video streams.

The parameters shall be implemented as a non-persistent Remote Shared Object available on the URI of the Flash Audio stream itself. The object shall be read-only for clients.

NOTE: Guidance is available from http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/net/SharedObject.html.

The object shall be named: **radiodns**

And have the following named string properties:

- **fqdn** for the **fqdn**
- **sid** for the **sid**

Clients shall listen for changes to these properties and update accordingly.

### 6.2.2 Metadata intervals

It is desirable that the client receives initial or updated parameters with as short a delay as possible. The cycle time of the metadata parameters will directly affect the speed at which connecting clients can access applications.

It is recommended that service providers ensure that connecting clients receive the parameters within 5 seconds.

## 6.3 Construction of bearerURI

The bearerURI for an IP-based service is constructed from the URL for the stream source.

EXAMPLE: `http://media-ice.musicradio.com/Capital`

## 6.4 Construction of ServiceIdentifier

The ServiceIdentifier for an IP-based service is constructed as follows:

`id/<fqdn>/<sid>`

# 7 Authoritative FQDN and ServiceIdentifier resolution from SPI SI

An Authoritative FQDN may also be provided in an SPI SI document [5]. In this case, the **fqdn** and **sid** parameters (see clause 6) are provided as attributes of the **radiodns** element:

- radiodns/@fqdn carries the **fqdn**
- radiodns/@serviceIdentifier carries the **sid**

EXAMPLE 1: `<radiodns fqdn="www.heart.co.uk" serviceIdentifier="bristol"/>`

The methods by which the SPI SI document can be acquired for the service are specified in ETSI TS 102 818 [5].

The ServiceIdentifier when derived from an SI document is constructed as follows:

- `id/<fqdn>/<sid>`

EXAMPLE 2: `id/www.heart.co.uk/bristol`

If the SPI SI document is either updated or expires through any applicable mechanism, the old parameters shall be discarded, and the process of resolving the Authoritative FQDN shall be repeated.

If the Authoritative FQDN has changed, then all active applications shall be notified and each application shall repeat its own process for connecting to resources using the updated Authoritative FQDN.

# 8 Implementation requirements

## 8.1 Service provider implementation

For broadcast services, a service provider shall support clause 5, Authoritative FQDN resolution for broadcast services. In addition, for services transmitted via FM with RDS/RBDS or DAB, the service provider shall transmit the ECC via RDS Group 1A or DAB FIG 0/9 respectively. For DAB or DRM services, the service provider may also support clause 7, Authoritative FQDN resolution from SPI SI. For IBOC services, the service provider shall transmit the Country Code and FCC Facility ID via the Station ID Number message (MSG ID = 0000). In the United States of America, the FCC Facility ID shall be populated by the FCC facility code of the transmitter. Outside of the United States of America, this shall be populated by a unique identifier supplied by the licensors of the IBOC technology.

For IP-streamed services, a service provider shall provide values for the **fqdn** and **ServiceIdentifier** parameters using at least one of the following:

- clause 6, Authoritative FQDN resolution for IP-streamed Services;

- clause 7, Authoritative FQDN resolution from SPI SI.

## 8.2 Device Implementation

For broadcast services, a device shall support clause 5, Authoritative FQDN resolution for broadcast services. For DAB or DRM services, the device may also support clause 7, Authoritative FQDN resolution from SPI SI.

For IP-streamed services, a device shall support the acquisition of values for the **fqdn** and **ServiceIdentifier** parameters from at least one of the following:

- clause 6, Authoritative FQDN resolution for IP-streamed Services;
- clause 7, Authoritative FQDN resolution from SPI SI.

# 9 Client identification

## 9.1 Introduction

A service provider's application may wish to identify connecting clients by using a client identifier. This enables it to provide a varying response to the client; for example, to provide more detailed or personalized responses to clients that present a client identifier, and less detailed or more generic responses to clients that do not.

The exact details on what this can be applied to are defined in the specification for that particular application.

The service provider's application may signal support for client identification by defining specific SRV records, as defined in the specification for that application.

The service provider shall return an error if it is determined that the presented client identifier is invalid or unrecognized, and shall always return a valid response if no device identifier is provided.

The method of allocation, distribution and management of client identifiers by a service provider is not defined by the present document.

Client identifiers shall be kept private to the client and shall not be made accessible to anything beyond that client, either through a user interface, or via any external connections.

## 9.2 Client identifier format

A client identifier is a string of between 16 and 128 characters, drawn from the allowed characters:

```
[a-zA-Z0-9]
```

It is strongly recommended that a client stores an acquired client identifier against the Authoritative FQDN(s) it has determined it applies to.

## 9.3 Client Identifier Presentation

### 9.3.1 Introduction

The common behaviour is defined as:

- A client shall determine whether the service provider's application supports the use of a client identifier, using the defined SRV records.
- A client may present a client identifier when supported by the service provider's application.
- A client shall only present the client identifier associated with the Authorative FQDN of the application it is requesting resources from, or connecting to.

- A client shall only present the client identifier for supported resources as given in the application specification. For all other resources, the client shall not present the client identifier.

### 9.3.2 Use of TLS

Implementation of the use of a client identifier shall require the use of Transport Layer Security (TLS) [13] encryption for any requests or connections that present the client identifier to an application and for any resulting responses or connections from the application to the client.

The appropriate version (or versions) of TLS will vary over time, based on the widespread deployment and known security vulnerabilities.

### 9.3.3 Protocol guidance

This clause contains specific guidance for a client to present the relevant client identifier over common protocols. In the present document, only a single protocol is defined: HTTPS.

Guidance for additional protocols specific to an application, or additional and more specific guidance may be given in the application specifications.

When requesting resources over HTTPS, a client shall present the relevant client identifier using the HTTP 1.1 authentication framework [14]. This defines a method of authentication for HTTP resources using an extensible set of challenge-response schemes.

The present document defines an additional authentication scheme to be used for the purposes of client identification, as follows:

In client presentation of credentials:

- the scheme name is **ClientIdentifier**;
- the credentials are defined in the "token68" syntax defined in IETF RFC 7235 [14] section 2.1, with the value computed from the client identifier string.

EXAMPLE 1: For example, to present a client identifier of **ABCD123456**, the client would use the following header field in the request for the resource:

```
Authorization: ClientIdentifier ABCD123456
```

When the service provider receives a request from the client with a valid client identifier, the service provider shall return an "HTTP 200 (OK)" response code, along with the resource contents in the response body.

EXAMPLE 2:

```
HTTP/1.1 200 OK
…
```

When the service provider receives a request from the client without a client identifier, the service provider shall return an "HTTP 401 (Unauthorized)" response code. This should also include the HTTP response header "WWW-Authenticate. The "WWW-Authenticate" response header indicates to the client that they are being challenged to provide a valid client identifier.

EXAMPLE 3:

```
HTTP/1.1 401 Unauthorized
WWW-Authenticate: ClientIdentifier
```

In challenges:

- the scheme name is **ClientIdentifier**;
- the authentication parameter **realm** is optional;
- the authentication parameter **charset** is optional;

- no other authentication parameters are defined. Unknown parameters shall be ignored by recipients, and new parameters can only be defined by revising the present document;
- both the scheme and the parameter names shall be matched case-insensitively.

When the service provider receives a request from the client with a client identifier that the service provider deems invalid or that is not recognised, the service provider shall return an "HTTP 403 (Forbidden)" response code.

EXAMPLE 4:

```
HTTP/1.1 403 Forbidden
```

On receiving this response, the client shall not repeat the same request.

## 9.4 Service provider reception

The common behaviour is defined as:

- a service provider's application shall ignore a presented client identifier if it does not intend to make use of it;
- a service provider's application shall return a valid response if the client identifier is not presented;
- a service provider shall signal that its application supports client identification through the use of SRV records determined by the application specification, should it wish to do this;
- a service provider's application shall return an error if the client identifier is invalid;
- a service provider's application shall return a valid response if the client identifier is valid;
- a service provider's application may vary its response based on a provided valid client identifier.

# Annex A (normative): Deriving the GCC for a service

## A.0 Introduction

The GCC is derived from the received ECC, see clause A.1. However, receivers need to cope with the situation that the ECC is not transmitted, or when the acquisition time is extended, see clause A.2.

## A.1 Deriving the GCC using ECC

The Global Country Code (GCC) shall be constructed by concatenating the Country Code and the Extended Country Code (ECC) of the service.

For FM RDS/RBDS or DAB/DAB+ audio services, the Country Code of the service is given by the **first** nibble of the RDS/RBDS PI code or DAB SId respectively. The ECC of the service is provided in the RDS Group 1A Block 3 Variant 0 [8] or the DAB FIG 0/9 [1] respectively.

For DAB data services, the Country Code of the service is given by the **third** nibble of the DAB SId. The ECC of the service is provided in the **first** and **second** nibbles of the DAB SId.

In all cases, the 1-character hexadecimal Country Code shall be combined with the 2-character hexadecimal ECC to create the 3-character hexadecimal GCC.

EXAMPLE 1: RDS PI code = C479, ECC = E1: GCC = C + E1 = CE1

EXAMPLE 2: DAB SId = D310, ECC = E0: GCC = D + E0 = DE0

EXAMPLE 3: DAB SId = E1F59B37: GCC = F + E1 = FE1

## A.2 Deriving the GCC without ECC

It is strongly recommended that receivers implement the process specified in figure A.1 using table A.1 and use it to derive the GCC if the ECC is not received promptly.

The process uses the Country Code of the service (derived from either the RDS/RBDS PI code or the DAB SId) and the ISO 3166-1 alpha-2 country code [10] of the current location of the receiver, and returns the GCC of the service. The process accounts for border areas, where a receiver may be located in a different country to the country of origination of the service.

Sources of information such as GPS/GSM/GeoIP can provide the ISO 3166-1 [10] alpha-2 country code for the **current location** of the receiver. The returned GCC shall be used when creating the RadioDNS FQDN and ServiceIdentifier.

Table A.1 has been constructed from several sources, which may change over time to reflect changing geo-political situations. The sources are:

- ISO 3166-1 alpha-2 Country Codes [10]
- Tables 3 to 7 in clause 5.4 of ETSI TS 101 756 [12]
- List of Countries and Territories by Land and Maritime Borders (Wikipedia: http://en.wikipedia.org/wiki/List_of_countries_and_territories_by_land_and_maritime_borders)

Table A.2 lists bordering countries that have been removed from table A.1, either because they are widely separated by water and it is unlikely that radio signals would cross between the countries, or because there is a very small and distant territorial presence claimed as part of the main territory (oceanic islands).

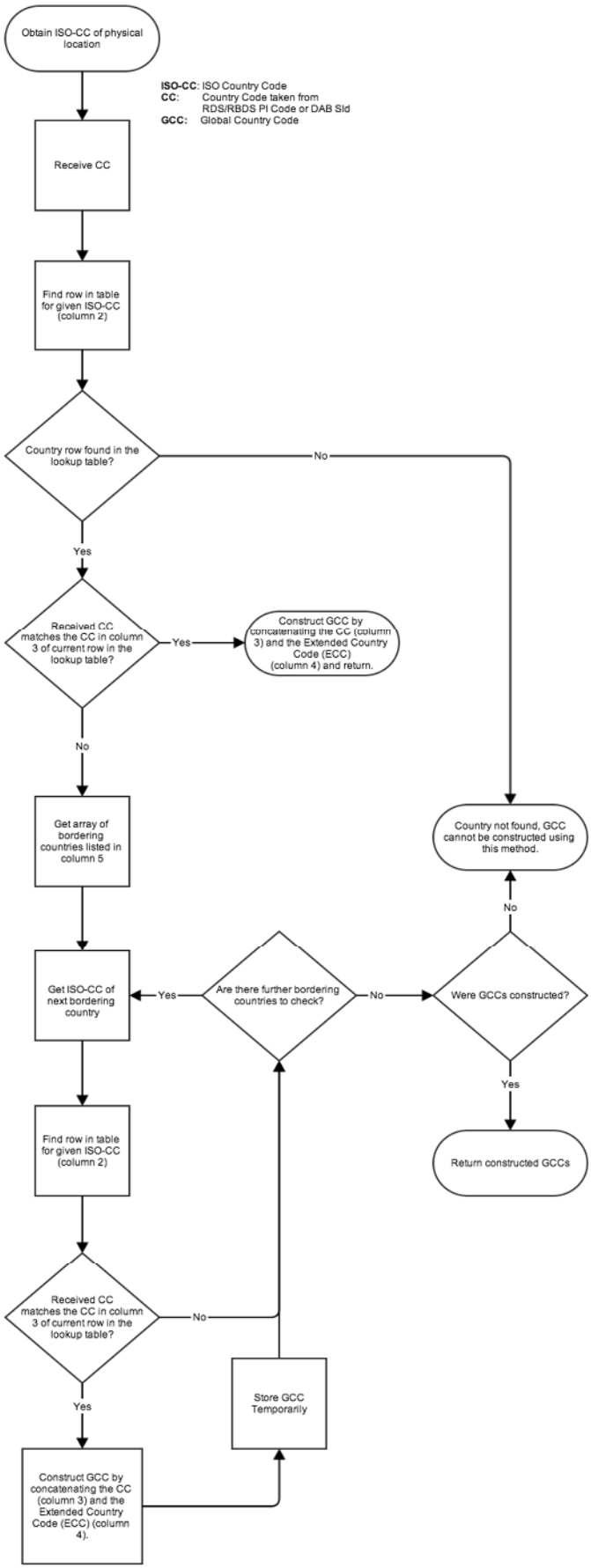



**Figure A.1: Process for GCC construction**

**Table A.1: Look-up table for GCC construction**

| Country | ISO 3166-1 [10] 2-alpha code | Country Code | ECC | Bordering countries look-up (CC:ISO 3166-1 [10] code) |
|---|---|---|---|---|
| Afghanistan | AF | A | F0 | C:CN; 8:IR; 4:PK; 5:TJ; E:TM; B:UZ |
| Albania | AL | 9 | E0 | C:HR; 1:GR; 5:IT; 3:MK; D:RS |
| Algeria | DZ | 2 | E0 | D:LY; 5:ML; 4:MR; 1:MA; 8:NE; E:ES; 7:TN; 3:EH |
| American Samoa | AS | X | XX | 4:WS; 3:TO |
| Andorra | AD | 3 | E0 | F:FR; E:ES |
| Angola | AO | 6 | D0 | C:CG; 1:NA; E:ZM |
| Anguilla | AI | 1 | A2 | 2:AG; 8:NL; F:VI |
| Antigua and Barbuda | AG | 2 | A2 | A:KN; 1:AI; 5:MS; F:FR |
| Argentina | AR | A | A2 | 1:BO; B:BR; C:CL; 6:PY; 9:UY; 4:FK |
| Armenia | AM | A | E4 | B:AZ; C:GE; 8:IR; 3:TR |
| Aruba | AW | 3 | A4 | B:DO; E:VE |
| Australia | AU | 1; 2; 3; 4; 5; 6; 7; 8 | F0 | C:ID; 9:PG; A:SB |
| Austria | AT | A | E0 | 2:CZ; D:DE; 1:DE; B:HU; 5:IT; 9:LI; 5:SK; 9:SI; 4:CH |
| Azerbaijan | AZ | B | E3 | A:AM; C:GE; 8:IR; 7:RU; 3:TR; E:TM |
| Bahamas | BS | F | A2 | 1:US; 2:US; 3:US; 4:US; 5:US; 6:US; 7:US; 8:US; 9:US; A:US; B:US; D:US; E:US |
| Bahrain | BH | E | F0 | 8:IR; 2:QA; 9:SA |
| Bangladesh | BD | 3 | F1 | B:MM; 5:IN |
| Barbados | BB | 5 | A2 | F:GY; C:VC; 6:TT; E:VE |
| Belarus | BY | F | E3 | 9:LV; C:LT; 8:PL; 7:RU; 6:UA |
| Belgium | BE | 6 | E0 | F:FR; D:DE; 1:DE; 7:LU; 8:NL; C:GB |
| Belize | BZ | 6 | A2 | 1:GT; 2:HN; F:MX |
| Benin | BJ | E | D0 | B:BF; 3:GH; 8:NE; F:NG; D:TG |
| Bermuda | BM | C | A2 | |
| Bhutan | BT | 2 | F1 | C:CN; 5:IN |
| Bolivia | BO | 1 | A3 | A:AR; B:BR; C:CL; 6:PY; 7:PE |
| Bosnia and Herzegovina | BA | F | E4 | C:HR; 1:ME; D:RS |
| Botswana | BW | B | D1 | 1:NA; A:ZA; E:ZM; 2:ZW |
| Brazil | BR | B | A2 | A:AR; 1:BO; 2:CO; F:GY; 6:PY; 7:PE; 8:SR; 9:UY; E:VE |
| British Indian Ocean Territory | IO | X | XX | B:MV |
| British Virgin Islands | VG | F | A5 | 8:PR; F:VI |
| Brunei | BN | B | F1 | F:MY |
| Bulgaria | BG | 8 | E1 | 1:GR; 3:MK; E:RO; D:RS; 3:TR |
| Burkina Faso | BF | B | D0 | E:BJ; C:CI; 3:GH; 5:ML; 8:NE; D:TG |
| Burma | MM | B | F0 | 3:BD; C:CN; 5:IN; 1:LA; 2:TH |
| Burundi | BI | 9 | D1 | 5:RW; D:TZ |
| Cambodia | KH | 3 | F2 | 1:LA; 2:TH; 7:VN |
| Cameroon | CM | 1 | D0 | 2:CF; 9:TD; C:CG; 7:GQ; 8:GA; F:NG |
| Canada | CA | C | A1 | 1:US; 2:US; 3:US; 4:US; 5:US; 6:US; 7:US; 8:US; 9:US; A:US; B:US; D:US; E:US; F:GL; F:PM |
| Cape Verde | CV | 6 | D1 | 8:GM; 4:MR; 7:SN |
| Cayman Islands | KY | 7 | A2 | 9:CU; 3:JM |
| Central African Republic | CF | 2 | D0 | 1:CM; 9:TD; C:CG; C:SD |
| Chad | TD | 9 | D2 | 1:CM; 2:CF; D:LY; 8:NE; F:NG; C:SD |
| Chile | CL | C | A3 | A:AR; 1:BO; 7:PE |
| China | CN | C | F0 | A:AF; 2:BT; B:MM; 5:IN; 9:JP; D:KZ; D:KP; 3:KG; 1:LA; F:MN; E:NP; 4:PK; 8:PH; 7:RU; 5:TJ; 7:VN; F:HK; 6:MO |
| Christmas Island | CX | X | XX | C:ID |
| Colombia | CO | 2 | A3 | B:BR; 8:CR; 3:EC; D:HT; 2:HN; 7:NI; 9:PA; 7:E:VE |
| Comoros | KM | C | D1 | F:FR; 4:MG; 3:MZ; B:SC; D:TZ |

| Country | ISO 3166-1 [10] 2-alpha code | Country Code | ECC | Bordering countries look-up (CC:ISO 3166-1 [10] code) |
|---|---|---|---|---|
| Democratic Republic of the Congo | CD | X | XX | 6:AO; 9:BI; 2:CF; C:CG; 5:RW; D:TZ; 4:UG; E:ZM |
| Republic of the Congo | CG | C | D0 | 6:AO; 1:CM; 2:CF; 8:GA |
| Cook Islands | CK | X | XX | 1:KI |
| Costa Rica | CR | 8 | A2 | 2:CO; 3:EC; 7:NI; 9:PA |
| Cote d'Ivoire | CI | C | D2 | B:BF; 3:GH; 9:GN; 2:LR; 5:ML |
| Croatia | HR | C | E3 | F:BA; B:HU; 5:IT; 1:ME; D:RS; 9:SI |
| Cuba | CU | 9 | A2 | D:HT; 2:HN; 3:JM; 7:KY |
| Curacao | CW | X | XX | B:DO; E:VE |
| Cyprus | CY | 2 | E1 | F:EG; 1:GR; 4:IL; A:LB; 3:TR |
| Czech Republic | CZ | 2 | E2 | A:AT; D:DE; 1:DE; 3:PL; 5:SK |
| Denmark | DK | 9 | E1 | D:DE; 1:DE; F:NO; 3:PL; E:SE; C:GB |
| Djibouti | DJ | 3 | D0 | E:ET; 7:SO; B:YE |
| Dominica | DM | A | A3 | F:FR; E:VE |
| Dominican Republic | DO | B | A3 | 2:CO; D:HT; 3:AW; CW; 8:PR; E:TC |
| Ecuador | EC | 3 | A2 | 2:CO; 8:CR; 7:PE |
| Egypt | EG | F | E0 | 2:CY; 1:GR; 4:IL; 5:JO; D:LY; 9:SA; C:SD; 3:TR |
| El Salvador | SV | C | A4 | 1:GT; 2:HN; 7:NI |
| Equatorial Guinea | GQ | 7 | D0 | 1:CM; 8:GA; F:NG |
| Eritrea | ER | X | XX | 3:DJ; 9:SA; C:SD; E:ET; B:YE |
| Estonia | EE | 2 | E4 | 6:FI; 9:LV; 7:RU; E:SE |
| Ethiopia | ET | E | D1 | 3:DJ; 6:KE; 7:SO; C:SD |
| Falkland Islands | FK | 4 | A2 | A:AR |
| Faroe Islands | FO | 9 | E1 | A:IS; F:NO; C:GB |
| Fiji | FJ | 5 | F1 | 9:NZ; 3:TO; F:VU |
| Finland | FI | 6 | E1 | 2:EE; F:NO; 7:RU; E:SE |
| France | FR | F | E1 | 3:AD; 6:BE; D:DE; 1:DE; 5:IT; 7:LU; B:MC; E:ES; 4:CH; C:GB |
| French Polynesia | PF | X | XX | 1:KI |
| Gabon | GA | 8 | D0 | 1:CM; C:CG; 7:GQ |
| The Gambia | GM | 8 | D1 | 6:CV; 7:SN |
| Georgia | GE | C | E4 | A:AM; B:AZ; 7:RU; 3:TR; 6:UA |
| Germany | DE | D;1 | E0 | A:AT; 6:BE; 2:CZ; 9:DK; F:FR; 7:LU; 8:NL; 3:PL; E:SE; 4:CH; C:GB |
| Ghana | GH | 3 | D1 | E:BJ; B:BF; C:CI; F:NG; D:TG |
| Gibraltar | GI | A | E1 | 1:MA; E:ES |
| Greece | GR | 1 | E1 | 9:AL; 8:BG; 2:CY; F:EG; 5:IT; D:LY; 3:MK; 3:TR |
| Greenland | GL | F | A1 | C:CA; A:IS; F:NO |
| Grenada | GD | D | A3 | C:VC; 6:TT |
| Guam | GU | X | XX | E:FM |
| Guatemala | GT | 1 | A4 | 6:BZ; C:SV; 2:HN; F:MX |
| Guernsey | GG | X | XX | F:FR; C:GB |
| Guinea | GN | 9 | D0 | C:CI; A:GW; 2:LR; 5:ML; 7:SN; 1:SL |
| Guinea Bissau | GW | A | D2 | 9:GN; 7:SN |
| Guyana | GY | F | A3 | 5:BB; B:BR; 8:SR; 6:TT; E:VE |
| Haiti | HT | D | A4 | F:BS; 2:CO; 9:CU; B:DO; 3:JM; E:TC |
| Honduras | HN | 2 | A4 | 6:BZ; 2:CO; 9:CU; C:SV; 1:GT; F:MX; 7:NI |
| Hong Kong | HK | F | F1 | |
| Hungary | HU | B | E0 | A:AT; C:HR; E:RO; D:RS; 5:SK; 9:SI; 6:UA |
| Iceland | IS | A | E2 | 9:FO; F:GL |
| India | IN | 5 | F2 | A:AF; 3:BD; 2:BT; B:MM; C:CN; E:NP; 4:PK; C:LK |
| Indonesia | ID | C | F2 | 1:AU; 2:AU; 3:AU; 4:AU; 5:AU; 6:AU; 7:AU; 8:AU; F:MY; 9:PG; A:SG |
| Iran | IR | 8 | F1 | A:AF; A:AM; B:AZ; B:IQ; 1:KW; 6:OM; 4:PK; 2:QA; 9:SA; 3:TR; E:TM; D:AE |
| Iraq | IQ | B | E1 | 8:IR; 5:JO; 1:KW; 9:SA; 3:TR |
| Ireland | IE | 2 | E3 | C:GB |
| Isle of Man | IM | X | XX | C:GB; 2:IE |
| Israel | IL | 4 | E0 | 2:CY; F:EG; 5:JO; A:LB |
| Italy | IT | 5 | E0 | 9:AL; 2:DZ; A:AT; C:HR; F:FR; 1:GR; D:LY; 3:SM; 9:SI; E:ES; 4:CH; 7:TN; 4:VA |
| Jamaica | JM | 3 | A3 | 2:CO; 9:CU; D:HT; 7:KY |

| Country | ISO 3166-1 [10] 2-alpha code | Country Code | ECC | Bordering countries look-up (CC:ISO 3166-1 [10] code) |
|---|---|---|---|---|
| Japan | JP | 9 | F2 | C:CN; E:KR; 8:PH; 7:RU |
| Jersey | JE | X | XX | F:FR; C:GB |
| Jordan | JO | 5 | E1 | F:EG; B:IQ; 4:IL; 9:SA |
| Kazakhstan | KZ | D | E3 | C:CN; 3:KG; 7:RU; E:TM; B:UZ |
| Kenya | KE | 6 | D2 | E:ET; 7:SO; D:TZ; 4:UG |
| Kiribati | KI | 1 | F1 | 7:NR |
| North Korea | KP | D | F0 | C:CN; 9:JP; E:KR; 7:RU |
| South Korea | KR | E | F1 | C:CN; 9:JP; D:KP |
| Kuwait | KW | 1 | F2 | 8:IR; B:IQ; 9:SA |
| Kyrgyzstan | KG | 3 | E4 | C:CN; D:KZ; 5:TJ; B:UZ |
| Laos | LA | 1 | F3 | B:MM; 3:KH; C:CN; 2:TH; 7:VN |
| Latvia | LV | 9 | E3 | F:BY; 2:EE; C:LT; 7:RU; E:SE |
| Lebanon | LB | A | E3 | 2:CY; 4:IL |
| Lesotho | LS | 6 | D3 | A:ZA |
| Liberia | LR | 2 | D1 | C:CI; 9:GN; 1:SL |
| Libya | LY | D | E1 | 2:DZ; 9:TD; F:EG; 1:GR; 5:IT; 8:NE; C:SD; 7:TN |
| Liechtenstein | LI | 9 | E2 | A:AT; 4:CH |
| Lithuania | LT | C | E2 | F:BY; 9:LV; 3:PL; 7:RU; E:SE |
| Luxembourg | LU | 7 | E1 | 6:BE; F:FR; D:DE; 1:DE |
| Macau | MO | 6 | F2 | |
| Republic of Macedonia | MK | 3 | E4 | 9:AL; 8:BG; 1:GR; D:RS |
| Madagascar | MG | 4 | D0 | C:KM; F:FR; 3:MZ; B:SC |
| Malawi | MW | F | D0 | 3:MZ; D:TZ; E:ZM |
| Malaysia | MY | F | F0 | B:BN; C:ID; 8:PH; A:SG; 2:TH; 7:VN |
| Maldives | MV | B | F2 | 5:IN; C:LK |
| Mali | ML | 5 | D0 | 2:DZ; B:BF; C:CI; 9:GN; 4:MR; 8:NE; 7:SN |
| Malta | MT | C | E0 | 5:IT; D:LY |
| Marshall Islands | MH | X | XX | 1:KI; E:FM; 7:NR |
| Mauritania | MR | 4 | D1 | 2:DZ; 6:CV; 5:ML; 1:MA; 7:SN; 3:EH |
| Mauritius | MU | A | D3 | F:FR; B:SC |
| Mayotte | YT | X | XX | C:KM; 4:MG |
| Mexico | MX | F | A4 | 6:BZ; 1:GT; 1:US; 2:US; 3:US; 4:US; 5:US; 6:US; 7:US; 8:US; 9:US; A:US; B:US; D:US; E:US |
| Federated States of Micronesia | FM | E | F3 | 9:PG |
| Moldova | MD | 1 | E4 | E:RO; 6:UA |
| Monaco | MC | B | E2 | F:FR |
| Mongolia | MN | F | F3 | C:CN; 7:RU |
| Montenegro | ME | 1 | E3 | 9:AL; F:BA; C:HR; 5:IT; D:RS |
| Montserrat | MS | 5 | A4 | 2:AG; F:FR; A:KN; E:VE |
| Morocco | MA | 1 | E2 | 2:DZ; 8:PT; E:ES; 4:MR; 3:EH |
| Mozambique | MZ | 3 | D2 | C:KM; 4:MG; F:MW; A:ZA; 5:SZ; D:TZ; E:ZM; 2:ZW |
| Namibia | NA | 1 | D1 | 6:AO; B:BW; A:ZA; E:ZM |
| Nauru | NR | 7 | F1 | 1:KI |
| Nepal | NP | E | F2 | 5:IN; C:CN |
| Netherlands | NL | 8 | E3 | 6:BE; D:DE; 1:DE; A:KN; C:GB; E:VE; 1:AI; F:VI |
| New Caledonia | NC | X | XX | 9:PG; A:SB; F:VU |
| New Zealand | NZ | 9 | F1 | |
| Nicaragua | NI | 7 | A3 | 8:CR; C:SV; 2:HN |
| Niger | NE | 8 | D2 | 2:DZ; E:BJ; B:BF; 9:TD; D:LY; 5:ML; F:NG |
| Nigeria | NG | F | D1 | E:BJ; 1:CM; 9:TD; 7:GQ; 3:GH; 8:NE |
| Niue | NU | X | XX | 3:TO |
| Norfolk Island | NF | X | XX | 9:NZ |
| Northern Mariana Islands | MP | X | XX | 9:JP |
| Norway | NO | F | E2 | 9:DK; 6:FI; A:IS; 7:RU; E:SE; C:GB; F:GL |
| Oman | OM | 6 | F1 | 8:IR; 4:P K; 9:SA; D:AE; B:YE |
| Pakistan | PK | 4 | F1 | A:AF; C:CN; 5:IN; 8:IR; 6:OM |
| Palau | PW | X | XX | C:ID; E:FM; 8:PH |
| Panama | PA | 9 | A3 | 2:CO; 8:CR |

| Country | ISO 3166-1 [10] 2-alpha code | Country Code | ECC | Bordering countries look-up (CC:ISO 3166-1 [10] code) |
|---|---|---|---|---|
| Papua New Guinea | PG | 9 | F3 | 1:AU; 2:AU; 3:AU; 4:AU; 5:AU; 6:AU; 7:AU; 8:AU; C:ID; E:FM; A:SB |
| Paraguay | PY | 6 | A3 | A:AR; 1:BO; B:BR |
| Peru | PE | 7 | A4 | 1:BO; B:BR; C:CL; 2:CO; 3:EC |
| Philippines | PH | 8 | F2 | C:ID; 9:JP; F:MY; 7:VN; D:TW |
| Poland | PL | 3 | E2 | F:BY; 2:CZ; 9:DK; D:DE; 1:DE; C:LT; 7:RU; 5:SK; E:SE; 6:UA |
| Portugal | PT | 8 | E4 | 1:MA; E:ES |
| Puerto Rico | PR | 8 | A3 | B:DO; E:VE; F:VG |
| Qatar | QA | 2 | F2 | E:BH; 8:IR; 9:SA; D:AE |
| Romania | RO | E | E1 | 8:BG; B:HU; 1:MD; D:RS; 3:TR; 6:UA |
| Russia | RU | 7 | E0 | B:AZ; F:BY; C:CN; 2:EE; 6:FI; C:GE; D:KZ; 9:LV; C:LT; F:MN; F:NO; 3:PL; E:SE; 6:UA; 1:US; 2:US; 3:US; 4:US; 5:US; 6:US; 7:US; 8:US; 9:US; A:US; B:US; D:US; E:US |
| Rwanda | RW | 5 | D3 | 9:BI; D:TZ; 4:UG |
| Saint Barthélemy | BL | X | XX | 2:AG; 8:NL; A:KN |
| Saint Helena Ascension and Tristan da Cunha | SH | A | D1 | |
| Saint Kitts and Nevis | KN | A | A4 | 2:AG; 8:NL; E:VE; 5:MS |
| Saint Lucia | LC | X | XX | 5:BB; F:FR; C:VC; E:VE |
| Saint Martin | MF | X | XX | 8:NL; 1:AI |
| Saint Pierre and Miquelon | PM | F | A6 | C:CA |
| Saint Vincent and the Grenadines | VC | C | A5 | 5:BB; D:GD; LC; 6:TT; E:VE |
| Samoa | WS | 4 | F2 | 3:TO |
| San Marino | SM | 3 | E1 | 5:IT |
| Saudi Arabia | SA | 9 | F0 | E:BH; F:EG; 8:IR; B:IQ; 5:JO; 1:KW; 6:OM; 2:QA; C:SD; D:AE; B:YE |
| Senegal | SN | 7 | D1 | 6:CV; 8:GM; 9:GN; A:GW; 5:ML; 4:MR |
| Serbia | RS | D | E2 | 9:AL; F:BA; 8:BG; C:HR; B:HU; 3:MK; 1:ME; E:RO |
| Seychelles | SC | B | A4 | C:KM; 4:MG; A:MU; D:TZ |
| Sierra Leone | SL | 1 | D2 | 9:GN; 2:LR |
| Singapore | SG | A | F2 | C:ID; F:MY |
| Slovakia | SK | 5 | E2 | A:AT; 2:CZ; B:HU; 3:PL; 6:UA |
| Slovenia | SI | 9 | E4 | A:AT; C:HR; 5:IT; B:HU |
| Solomon Islands | SB | A | F1 | 1:AU; 2:AU; 3:AU; 4:AU; 5:AU; 6:AU; 7:AU; 8:AU; 9:PG; F:VU |
| Somalia | SO | 7 | D2 | 3:DJ; E:ET; 6:KE; B:YE |
| South Africa | ZA | A | D0 | B:BW; 6:LS; 3:MZ; 1:NA; 5:SZ; 2:ZW |
| South Sudan | SS | X | XX | 2:CF; E:ET; 6:KE; C:SD; 4:UG |
| Spain | ES | E | E2 | 2:DZ; 3:AD; F:FR; 5:IT; 1:MA; 8:PT; A:GI |
| Sri Lanka | LK | C | F1 | 5:IN; B:MV |
| Sudan | SD | C | D3 | 2:CF; 9:TD; F:EG; E:ET; D:LY |
| Suriname | SR | 8 | A4 | B:BR; F:FR; F:GY |
| Svalbard | SJ | X | XX | 7:RU; F:GL |
| Swaziland | SZ | 5 | D2 | 3:MZ; A:ZA |
| Sweden | SE | E | E3 | 9:DK; 2:EE; 6:FI; D:DE; 1:DE; C:LT; F:NO; 3:PL; 7:RU |
| Switzerland | CH | 4 | E1 | A:AT; F:FR; 5:IT; 9:LI; D:DE; 1:DE |
| Taiwan | TW | D | F1 | C:CN; 9:JP; 8:PH |
| Tajikistan | TJ | 5 | E3 | A:AF; C:CN; 3:KG; B:UZ |
| Tanzania | TZ | D | D1 | 9:BI; C:KM; 6:KE; F:MW; 3:MZ; 5:RW; B:SC; 4:UG; E:ZM |
| Thailand | TH | 2 | F3 | B:MM; 3:KH; 5:IN; C:ID; 1:LA; F:MY; 7:VN |
| Togo | TG | D | D0 | E:BJ; B:BF; 3:GH |
| Tokelau | TK | X | XX | 1:KI; 4:WS |
| Tonga | TO | 3 | F3 | 5:FJ; 9:NZ; 4:WS |
| Trinidad and Tobago | TT | 6 | A4 | 5:BB; D:GD; F:GY; E:VE |
| Tunisia | TN | 7 | E2 | 2:DZ; 5:IT; D:LY |

| Country | ISO 3166-1 [10] 2-alpha code | Country Code | ECC | Bordering countries look-up (CC:ISO 3166-1 [10] code) |
|---|---|---|---|---|
| Turkey | TR | 3 | E3 | A:AM; B:AZ; 8:BG; 2:CY; F:EG; C:GE; 1:GR; 8:IR; B:IQ; E:RO; 7:RU; 6:UA |
| Turkmenistan | TM | E | E4 | A:AF; 8:IR; D:KZ; B:UZ |
| Turks and Caicos Islands | TC | E | A3 | F:BS; B:DO; D:HT |
| Tuvalu | TV | X | XX | 5:FJ; 1:KI |
| Uganda | UG | 4 | D2 | 6:KE; 5:RW; SS; D:TZ |
| Ukraine | UA | 6 | E4 | F:BY; B:HU; C:GE; 1:MD; 3:PL; E:RO; 7:RU; 5:SK; 3:TR |
| United Arab Emirates | AE | D | F2 | 8:IR; 6:OM; 2:QA; 9:SA |
| United Kingdom | GB | C | E1 | 6:BE; 9:DK; F:FR; D:DE; 1:DE; 2:IE; 8:NL |
| United States | US | 1;2;3;4;5;6;7;8;9;A;B;D;E | A0 | C:CA; 9:CU; 1:KI; F:MX; 7:RU |
| United States Virgin Islands | VI | F | A5 | 8:NL; E:VE; 1:AI; F:VG |
| Uruguay | UY | 9 | A4 | A:AR; B:BR |
| Uzbekistan | UZ | B | E4 | A:AF; D:KZ; 3:KG; 5:TJ; E:TM |
| Vanuatu | VU | F | F2 | 5:FJ; A:SB |
| Vatican City | VA | 4 | E2 | 5:IT |
| Venezuela | VE | E | A4 | 5:BB; B:BR; 2:CO; A:DM; F:GY; 8:NL; C:VC; 6:TT; 3:AW; 8:PR |
| Vietnam | VN | 7 | F2 | 3:KH; C:CN; C:ID; 1:LA; F:MY; 8:PH; 2:TH |
| Wallis and Futuna | WF | X | XX | 5:FJ; 4:WS; 3:TO |
| Western Sahara | EH | 3 | D3 | 2:DZ; 4:MR; 1:MA; E:ES |
| Yemen | YE | B | F3 | 3:DJ; 6:OM; 9:SA; 7:SO |
| Zambia | ZM | E | D2 | 6:AO; B:BW; F:MW; 3:MZ; 1:NA; D:TZ; 2:ZW |
| Zimbabwe | ZW | 2 | D2 | B:BW; 3:MZ; A:ZA; E:ZM |
| NOTE: An "X" in column 3 and "XX" in column 4 indicate that no codes are allocated for this country and the broadcast is registered in an adjoining country. | | | | |

**Table A.2: Countries considered non-bordering for the purposes of GCC construction**

| Country | ISO 3166-1 [10] 2-alpha code | Bordering Country removed from table A.1 | Reason for removal |
|---|---|---|---|
| Albania | AL | Montenegro | > 200 km separation over land |
| Algeria | DZ | Italy | > 200 km separation over sea |
| Australia | AU | New Zealand | > 200 km separation over sea |
| Bahamas | BS | Cuba | > 200 km separation over sea |
| Bahamas | BS | Haiti | > 200 km separation over sea |
| Bahamas | BS | Turks and Caicos Islands | > 200 km separation over sea |
| Barbados | BB | France | > 200 km separation over sea |
| Brazil | BR | France | > 200 km separation over sea |
| Cayman Islands | KY | Colombia | > 200 km separation over sea |
| Cayman Islands | KY | Honduras | > 200 km separation over sea |
| China | CN | Taiwan | > 200 km separation over sea |
| China | CN | South Korea | > 200 km separation over sea |
| Colombia | CO | Dominican Republic | > 200 km separation over sea |
| Colombia | CO | Jamaica | > 200 km separation over sea |
| Colombia | CO | Peru | > 200 km separation over sea |
| Colombia | CO | Cayman Islands | > 200 km separation over sea |
| Cuba | CU | Bahamas | > 200 km separation over sea |
| Cuba | CU | Mexico | > 200 km separation over sea |
| Cuba | CU | United States | > 200 km separation over sea |
| Dominican Republic | DO | Venezuela | > 200 km separation over sea |
| Honduras | HN | Cayman Islands | > 200 km separation over sea |
| India | IN | Thailand | > 200 km separation over sea |
| India | IN | Maldives | > 200 km separation over sea |
| India | IN | Indonesia | > 200 km separation over sea |
| Indonesia | ID | Thailand | > 200 km separation over sea |
| Indonesia | ID | India | > 200 km separation over sea |
| Indonesia | ID | Vietnam | > 200 km separation over sea |
| Indonesia | ID | Philippines | > 200 km separation over sea |
| Iran | IR | Bahrain | > 200 km separation over sea |
| Italy | IT | Malta | > 100 km separation over sea |
| Italy | IT | Montenegro | > 200 km separation over sea |
| Japan | JP | North Korea | > 200 km separation over sea |
| Japan | JP | Taiwan | > 200 km separation over sea |
| Libya | LY | Malta | > 200 km separation over sea |
| Mexico | MX | Cuba | > 200 km separation over sea |
| Mexico | MX | Honduras | > 200 km separation over sea |
| Mozambique | MZ | France | > 200 km separation over sea |
| New Caledonia | NC | Australia | > 200 km separation over sea |
| New Caledonia | NC | Fiji | > 200 km separation over sea |
| New Zealand | NZ | Australia | > 200 km separation over sea |
| New Zealand | NZ | Tonga | > 200 km separation over sea |
| New Zealand | NZ | Fiji | > 200 km separation over sea |
| Nicaragua | NI | Colombia | > 200 km separation over sea |
| Norway | NO | Faroe Islands | > 200 km separation over sea |
| Philippines | PH | China | > 200 km separation over sea |
| Russia | RU | Japan | > 200 km separation over sea |
| Russia | RU | North Korea | > 200 km separation over sea |
| Russia | RU | Turkey | > 200 km separation over sea |
| Spain | ES | Western Sahara | > 100 km separation over sea |
| Sudan | SD | Saudi Arabia | > 200 km separation over sea |
| Sweden | SE | Latvia | > 100 km separation over sea |
| Turkmenistan | TM | Azerbaijan | > 200 km separation over sea |
| United Kingdom | GB | Faroe Islands | > 200 km separation over sea |
| United Kingdom | GB | Norway | > 200 km separation over sea |
| United States | US | Bahamas | > 100 km separation over sea |
| Venezuela | VE | Montserrat | > 200 km separation over sea |
| Venezuela | VE | Dominican Republic | > 200 km separation over sea |
| Venezuela | VE | Saint Kitts and Nevis | > 200 km separation over sea |
| Venezuela | VE | France | > 200 km separation over sea |
| Venezuela | VE | United States Virgin Islands | > 200 km separation over sea |

# Annex B (informative): Bibliography


- Microsoft™ Corporation (Revision 01.20.06, January 2012): "Advanced Systems Format (ASF) Specification".

# History

| Document history | | |
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| V1.3.1 | May 2019 | Publication |
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